

Week 2: Classification Fundamentals - Pre-Class Study Guide

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Time Required: 30 minutes **Purpose:** Prime your understanding of classification concepts before the live session

Welcome to Week 2!

Last week you mastered **regression** - predicting continuous values like house prices. This week we're tackling **classification** - predicting categories like "yes/no", "positive/negative", or "benign/malignant".

The good news: The sklearn API pattern you learned (fit/predict) stays exactly the same!

The new skills: Understanding different evaluation metrics and learning when accuracy can lie to you.

Pre-Class Activities (Complete Before Live Session)

Part 1: Watch These Videos (25 minutes total)

Watch in this order:

1. **StatQuest - Logistic Regression** (15 min)

[Watch: StatQuest - Logistic Regression](#)

What to focus on:

- How logistic regression creates an S-shaped curve
- Why it's called "regression" even though it does classification
- What probability outputs mean

2. **StatQuest - Confusion Matrix** (8 min)

[Watch: StatQuest - Confusion Matrix](#)

What to focus on:

- The 2x2 grid structure (you'll draw this many times today!)
- True Positives, True Negatives, False Positives, False Negatives
- Why this matters for understanding model performance

3. **Optional: Google ML Crash Course - Classification** (Read sections 1-3 only)

[Click here for Google ML Crash Course - Classification](#)

What to focus on:

- Decision boundaries
- Why classification is different from regression

Part 2: Key Concepts to Understand

By the end of the videos, you should have a basic understanding of:

Classification vs Regression

Regression	Classification
Predicts numbers	Predicts categories
Example: House price (\$250,000)	Example: Spam or not spam
Outputs: Continuous values	Outputs: Class labels (0 or 1)
Week 1 focus	Week 2 focus

Binary Classification

Binary means two classes: - Yes / No - Positive / Negative - Malignant / Benign (cancer vs not cancer) - Fraud / Legitimate

This week's dataset: Breast Cancer Wisconsin - **Class 0 (malignant):** Cancer is present - **Class 1 (benign):** No cancer

The Confusion Matrix (Preview)

You'll become very familiar with this 2x2 grid:

		Predicted	
		Neg	Pos
Actual	Neg	TN	FP
	Pos	FN	TP

Don't worry if this looks confusing! We'll spend significant time on this during the live session. For now, just notice the structure.

Logistic Regression Basics

Key points from the video:

- Takes continuous inputs (like linear regression)
- Squashes output through an S-shaped curve (sigmoid function)
- Outputs probabilities between 0 and 1
- Converts probabilities to class predictions using a threshold (default: 0.5)

Analogy: Think of it like a dimmer switch: - Linear regression: outputs any brightness (0 to infinity watts)
- Logistic regression: outputs brightness percentage (0% to 100%)

Part 3: Vocabulary Pre-Reading

These terms will appear frequently in Week 2. You don't need to memorize them now - just get familiar:

Term	Quick Definition
Binary Classification	Predicting one of two classes (0 or 1)
Sigmoid Function	S-shaped curve that converts any number to 0-1 range
Probability Threshold	Cutoff point (default 0.5) for converting probability to class
Confusion Matrix	2x2 table showing correct and incorrect predictions
True Positive (TP)	Correctly predicted positive class
True Negative (TN)	Correctly predicted negative class
False Positive (FP)	Incorrectly predicted positive (false alarm)
False Negative (FN)	Incorrectly predicted negative (missed case)
Precision	Of all positive predictions, how many were correct?
Recall	Of all actual positives, how many did we catch?
Accuracy	Percentage of all predictions that were correct

Medical test analogy: Think of a COVID test: - **TP:** Test says positive, you have COVID - **TN:** Test says negative, you don't have COVID - **FP:** Test says positive, but you're healthy (false alarm) - **FN:** Test says negative, but you have COVID (missed diagnosis!)

Part 4: Connection to Week 1

What stays the same:

- Load data with sklearn
- Split into training and test sets (BEFORE preprocessing!)
- Use `model.fit(X_train, y_train)` to train
- Use `model.predict(X_test)` to make predictions
- Evaluate performance with metrics

What's new:

- Using `LogisticRegression` instead of `LinearRegression`
- New method: `predict_proba()` for probability outputs
- Different metrics: confusion matrix, precision, recall instead of MSE/RMSE
- Binary target variable (0 or 1) instead of continuous

The pattern is the same. The tools are the same. Only the problem type changed!

Part 5: Questions to Think About

Come to the live session with thoughts on these questions:

1. If a cancer screening test is 99% accurate, does that mean it's a great test? (Hint: Think about what happens if only 1% of people have cancer...)
2. In spam filtering, which is worse:
 - Letting spam through to your inbox? (False Negative)
 - Blocking an important email? (False Positive)
3. For cancer diagnosis, which is worse:
 - Telling a healthy person they have cancer? (False Positive)
 - Missing a cancer diagnosis? (False Negative)

These questions have no universal "right" answer - it depends on context! That's why we need multiple metrics, not just accuracy.

Part 6: Pre-Class Practice (Optional but Recommended)

If you have an extra 10-15 minutes, try this quick exercise:

Draw a confusion matrix from memory: 1. Get a piece of paper 2. Draw a 2x2 grid 3. Label it with: Actual (rows), Predicted (columns), TP, TN, FP, FN 4. Check your drawing against the visual in Part 2

Try a simple calculation:

Imagine a cancer test with these results on 100 patients: - 90 healthy patients, 10 with cancer - Test correctly identifies 8 cancer cases (TP = 8) - Test misses 2 cancer cases (FN = 2) - Test correctly identifies 85 healthy (TN = 85) - Test gives 5 false alarms (FP = 5)

Calculate: - Accuracy = $(TP + TN) / \text{Total} = ?$ - What percentage of cancer cases did we catch? (This is called “recall”)

Answers: - Accuracy = $(8 + 85) / 100 = 93\%$ - Recall = $8 / 10 = 80\%$ (we caught 80% of cancer cases)

What to Expect in Week 2

Week 2 has TWO sessions:

Day 1: Classification Fundamentals (3 hours)

Schedule: - 0:00-0:15: Week 1 recap & Week 2 overview - 0:15-0:40: Classification concepts (confusion matrix on whiteboard) - 0:40-1:00: Logistic regression theory - 1:00-1:15: Advanced classification concepts - 1:15-1:30: **BREAK** - 1:30-2:25: Live coding - complete classification pipeline (Breast Cancer dataset) - 2:25-2:35: Threshold tuning interpretation - 2:35-2:45: Business cost analysis - 2:45-3:15: Pair programming - Titanic survival prediction - 3:15-3:20: Wrap-up & preview Day 2

What to bring: - Questions from the videos - Notebook for taking notes during whiteboard session - Curiosity about metrics!

Day 2: Deep Dive & Guided Practice (1 hour)

Schedule: - 0:00-0:05: Recap - 0:05-0:25: Deep dive on class imbalance and metric selection - 0:25-0:45: Guided practice on new dataset (Heart Disease) with instructor support - 0:45-0:55: Quiz (8 questions, 60% to pass) - 0:55-1:00: Week 3 preview

Purpose: Day 2 gives you deeper understanding and more practice with instructor support before tackling homework solo.

Success Checklist

Before the live session, you should:

- ☐ Watched StatQuest - Logistic Regression video
 - ☐ Watched StatQuest - Confusion Matrix video
 - ☐ Reviewed vocabulary list
 - ☐ Thought about the discussion questions
 - ☐ (Optional) Practiced drawing confusion matrix
 - ☐ Ready to learn about classification!
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If You're Feeling Overwhelmed

Don't worry! These concepts are new and can feel abstract. The live session is designed to make everything concrete through:

- Visual demonstrations
- Live coding examples
- Real dataset (cancer screening)
- Pair programming practice
- Lots of repetition

Remember: You don't need to master these concepts before class. The pre-work is meant to prime your brain, not teach you everything. The real learning happens during the live session!

Questions Before Class?

If anything is unclear, don't stress! The live session will bring it all together.

Common questions we'll answer:

- "Why is it called logistic REGRESSION if it does classification?"
 - "How do I choose between precision and recall?"
 - "When is accuracy misleading?"
 - "What's the difference between `predict()` and `predict_proba()`?"
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See you in the live session!

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